OPTIMA: A Modular Vision-to-Speech Glasses Prototype

Reference Circuit Design and Analytical Feasibility Study

Prototype design paper prepared in the Hacktoberfest26 Rabat context

Abstract

Wearable visual assistance can make printed text and nearby scenes easier to interpret, but the practical usefulness of a prototype depends on image quality, accessible feedback, power integrity, and a reliable path from camera capture to spoken output. This paper presents OPTIMA, a modular design in which an ESP32-S3 camera node acquires a still image, a hosted open-weight vision-language model generates a task-specific response, and a compatible companion web app will present the result as text and speech. The study combines prior research on visual questions from blind and low-vision users with manufacturer pin maps and component datasheets. It specifies a core capture circuit and optional haptic, digital-audio, and battery configurations. Standard electrical schematics are paired with color-coded assembly illustrations and an explicit breadboard connection schedule. Analytical calculations establish example power and timing budgets, while a proposed evaluation protocol separates answer correctness, useful abstention, latency, and electrical reliability. The contribution is a reproducible reference design and validation method; hardware performance and user benefit have not yet been experimentally established.

Keywords: assistive technology; smart glasses; visual question answering; ESP32-S3; vision-to-speech; open-weight AI; reference circuits.

1. Introduction

An assistive glasses prototype should convert a deliberate user request into useful, recoverable feedback. A camera alone is insufficient: the system also needs a capture control, an observable acquisition state, understandable speech, and a mechanism for handling images that cannot support an answer. OPTIMA targets stationary reading and scene interpretation. The same interface can also support translation and explanation for general users.

The central design choice is to separate sensing from model inference. The frame-mounted node performs acquisition and communication; the backend owns model access; and the companion web app will handle interaction and speech on a phone or laptop. This reduces the hardware scope and keeps model credentials away from the wearable.

1.1. Related work and motivation

The VizWiz study demonstrates that photographs and questions supplied by blind users differ from curated visual-question-answering inputs: image quality is variable and some questions are unanswerable [1]. Subsequent work on long-form answers reports plausible descriptions containing incorrect visual details, particularly for insufficient images [2]. These findings motivate explicit recapture feedback and evaluation of unsupported claims.

OpenGlass provides a hardware precedent based on a XIAO ESP32-S3 Sense and a removable glasses mount; its repository now directs development to Omi [3]. OPTIMA adopts the modular sensing principle while specifying a distinct task-oriented interface and reference circuit set. Neither that precedent nor this design establishes navigation safety or clinical effectiveness.

2. System Architecture and Design Method

A distributed capture-to-speech path with explicit responsibilities and validation boundaries.

The planned platform couples the XIAO ESP32-S3 Sense glasses model with an OPTIMA web app on a phone or laptop, the existing FastAPI vision function, and a hosted Gemma model. On request, the glasses will upload one image for reading, scene description, translation, or explanation [17].

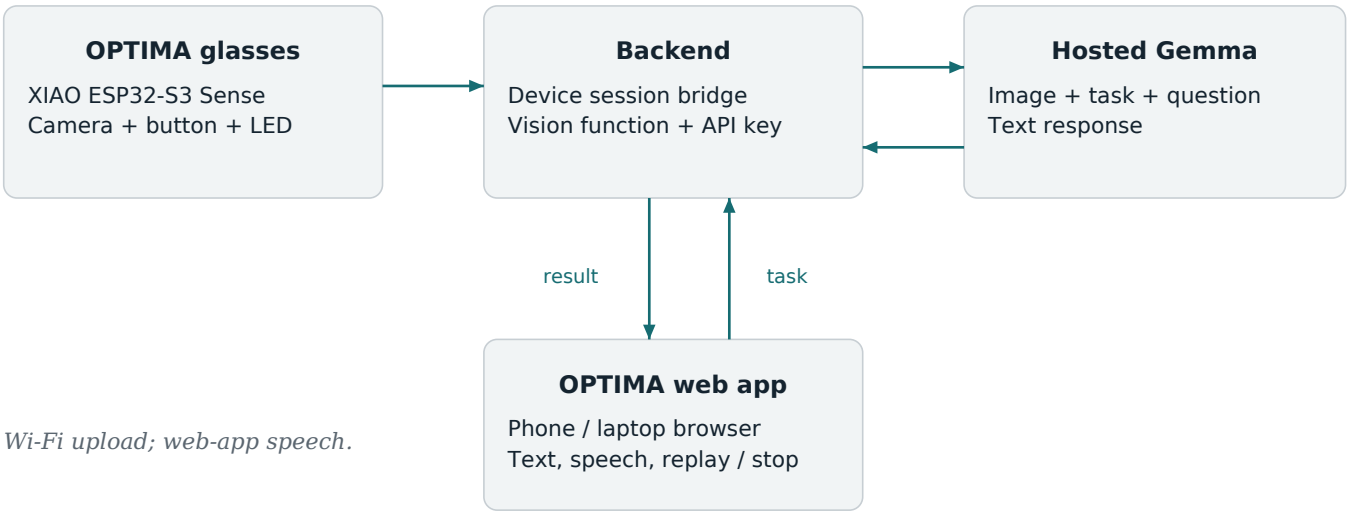


Figure 1. Proposed architecture for OPTIMA glasses and the companion web app. The app will exchange task settings and results with the backend. A device/session bridge will connect camera uploads to the paired browser session.

2.1. Source-based design procedure

The electrical design was derived by cross-checking the board pin map, base-board schematic, expansion-board schematic, and peripheral documentation [4]-[9]. External controls were allocated to GPIOs that do not occupy the camera, microphone, or SD-card signals. Optional audio uses explicitly assigned I2S outputs, rather than assuming that default peripheral pins are available.

Separate names identify the control rail, auxiliary motor rail, USB supply, and battery input. Component choices are checked against voltage, current, drive-level, and output topology requirements. Manufacturer figures and the analytical examples below are not substitutes for measurements on the purchased revision.

2.2. Web app and glasses compatibility

An OPTIMA web app will be developed for the selected XIAO ESP32-S3 Sense glasses model. It will offer task selection, text results, speech playback, replay, and stop controls in a phone or laptop browser. Compatibility will use Wi-Fi image uploads to a shared backend and a paired device session.

The current POST /analyze endpoint replies only to its caller. A session bridge will publish each result with an identifier; the web app will poll or subscribe and speak it once. Browser and board-revision compatibility will be tested during integration. Bluetooth is not required; browser support is not universal [18]. Model keys will remain on the server.

3. Core Capture and User-Feedback Circuit

The minimum electronics needed for deliberate image acquisition and visible capture status.

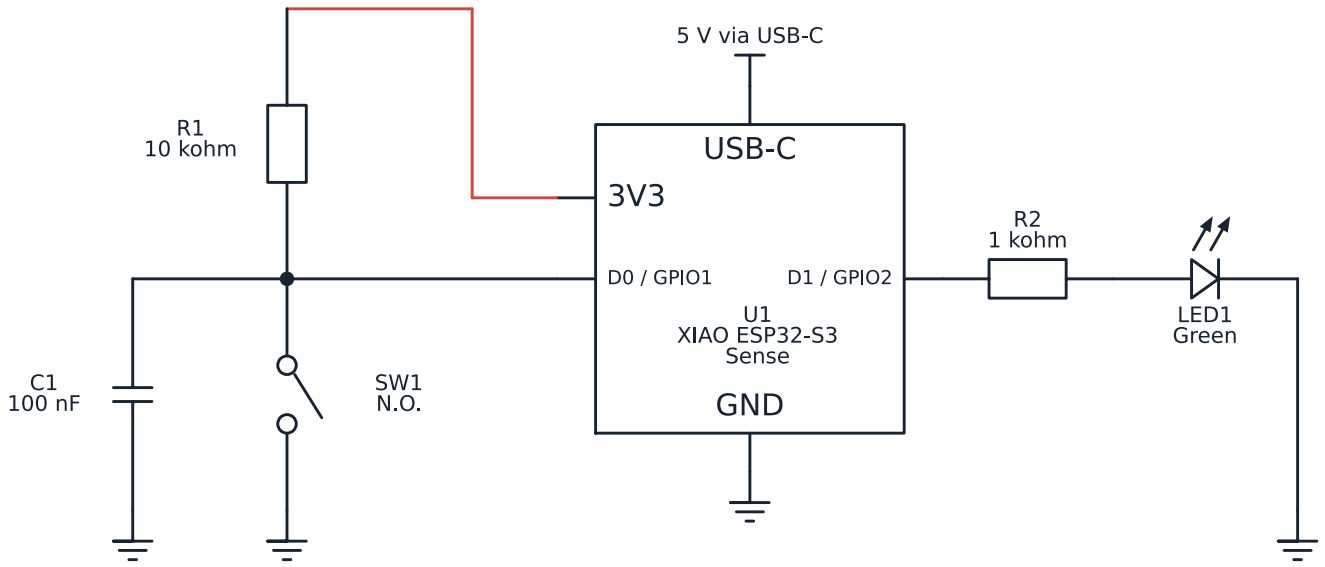


Figure 2. Electrical schematic of the core node. SW1 is normally open. R1 and C1 condition the capture input; R2 limits the current through LED1. The supplied Sense camera assembly remains on its expansion connector.

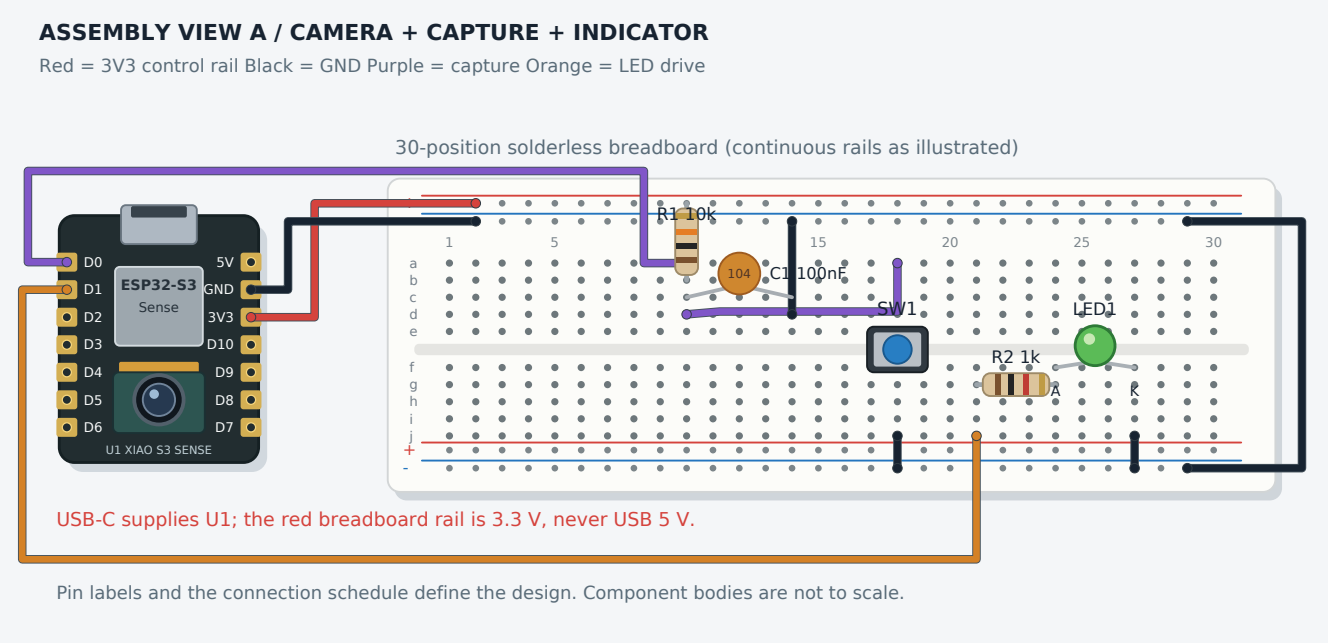


Figure 3. Color-coded breadboard assembly view of Figure 2. A red rail carries 3V3 control power; black wires connect the two ground rails. Appendix A specifies terminal locations and tie-group connectivity. Bodies are illustrative, not to scale.

Remark 1 (Input conditioning). *R1 = 10 kohm and C1 = 100 nF give an RC time constant of 1 ms. A separate firmware debounce interval of approximately 30 ms is a starting design choice. The indicator should turn on before acquisition, and repeated presses should be ignored while a request is in progress.*

4. Optional Haptic Feedback Circuit

A MOSFET-switched ERM motor on an auxiliary regulated supply.

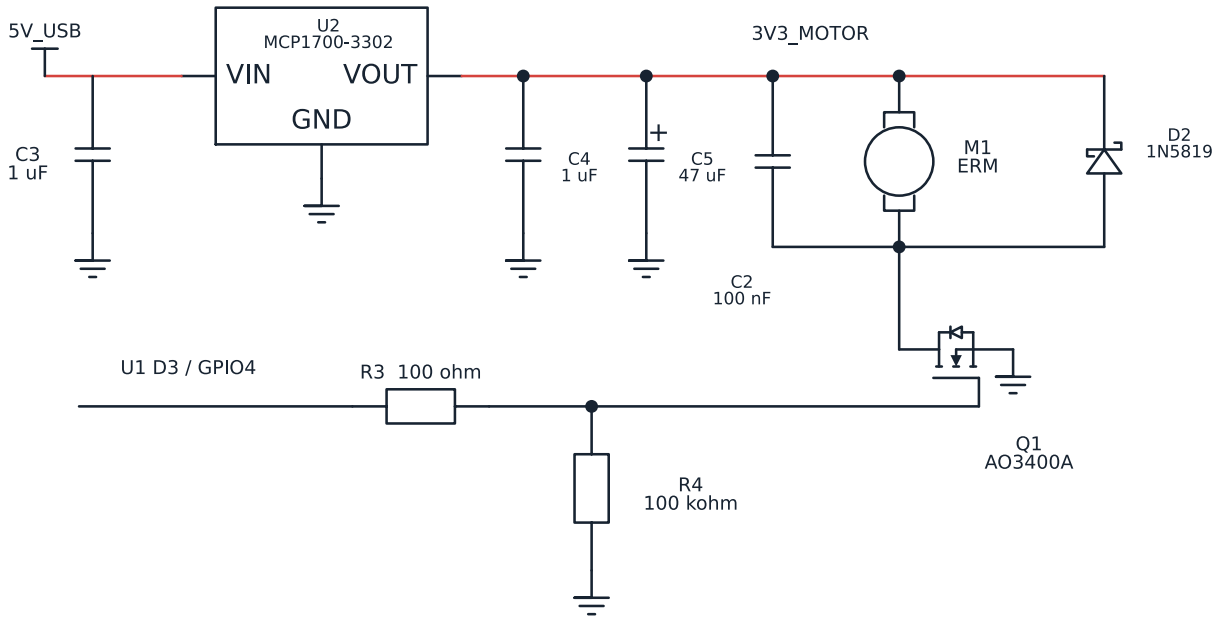


Figure 4. Motor-driver reference schematic. D2 is reverse-biased during normal operation (cathode to 3V3_MOTOR) and clamps the motor inductive transient. Q1 is a logic-driven N-channel MOSFET; R4 holds the gate low during reset.

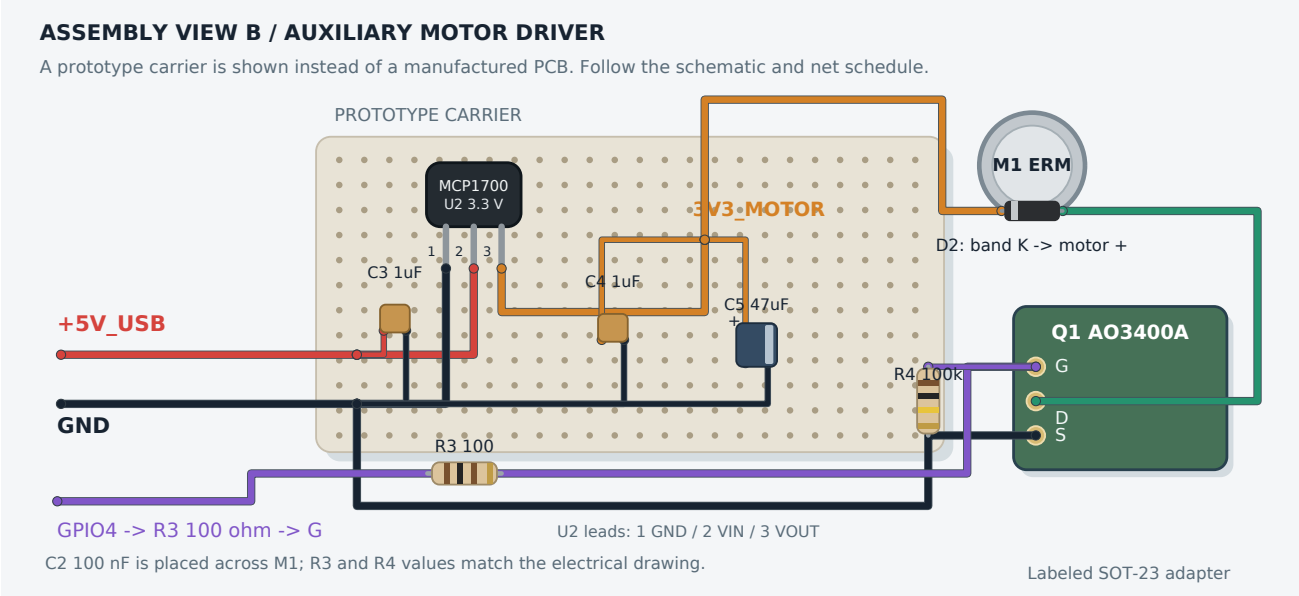


Figure 5. Part-oriented assembly illustration. The carrier is a functional arrangement of named nets, not a fabricated PCB. Use the labeled Q1 adapter terminals and add C2 at the motor terminals as specified in Figure 4.

Remark 2 (Supply and switch selection). *AO3400A is selected using its on-resistance rating at 2.5 V gate drive, not its threshold voltage [12]. The motor is specified for 2-5 V, with 60 mA listed at 3 V [14]; its startup current must be checked against the MCP1700 250 mA rating [13]. At an assumed 80 mA, regulator dissipation is (5 - 3.3) x 0.08 = 0.136 W. Share ground, but do not join 3V3_MOTOR to U1 3V3.*

5. Optional I2S Audio Playback

Optional local speaker path; the initial wearable will use speech from the web app.

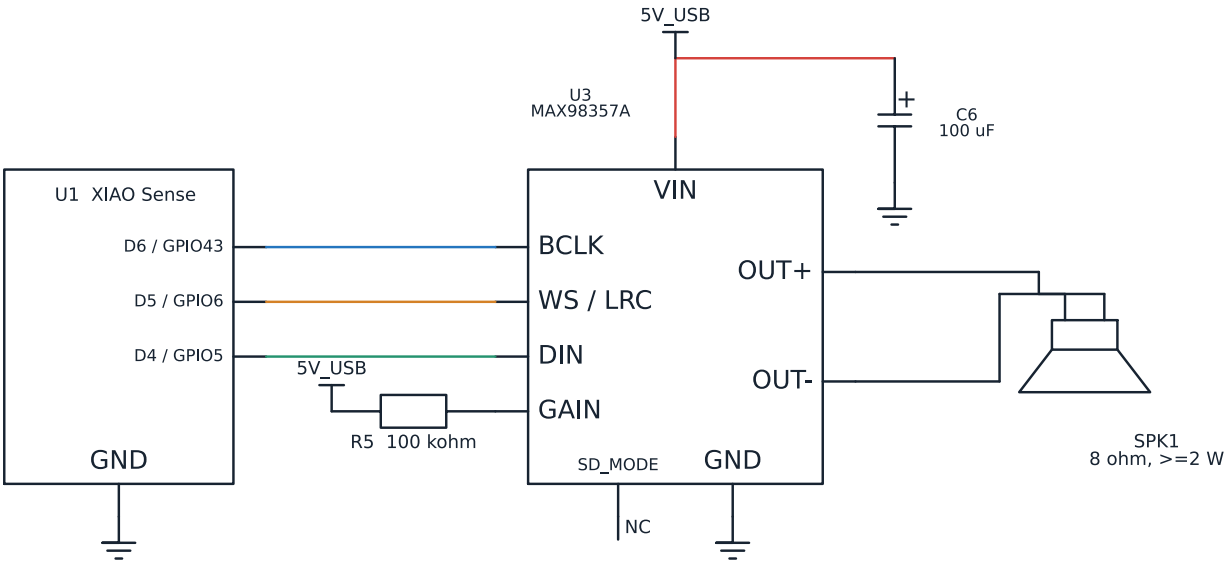


Figure 6. Digital audio reference schematic using a MAX98357A breakout. Proposed signals are BCLK = GPIO43, WS = GPIO6, and DOUT = GPIO5. R5 selects the specified breakout's 3 dB gain setting; the onboard SD bias is retained [10], [11], [19].

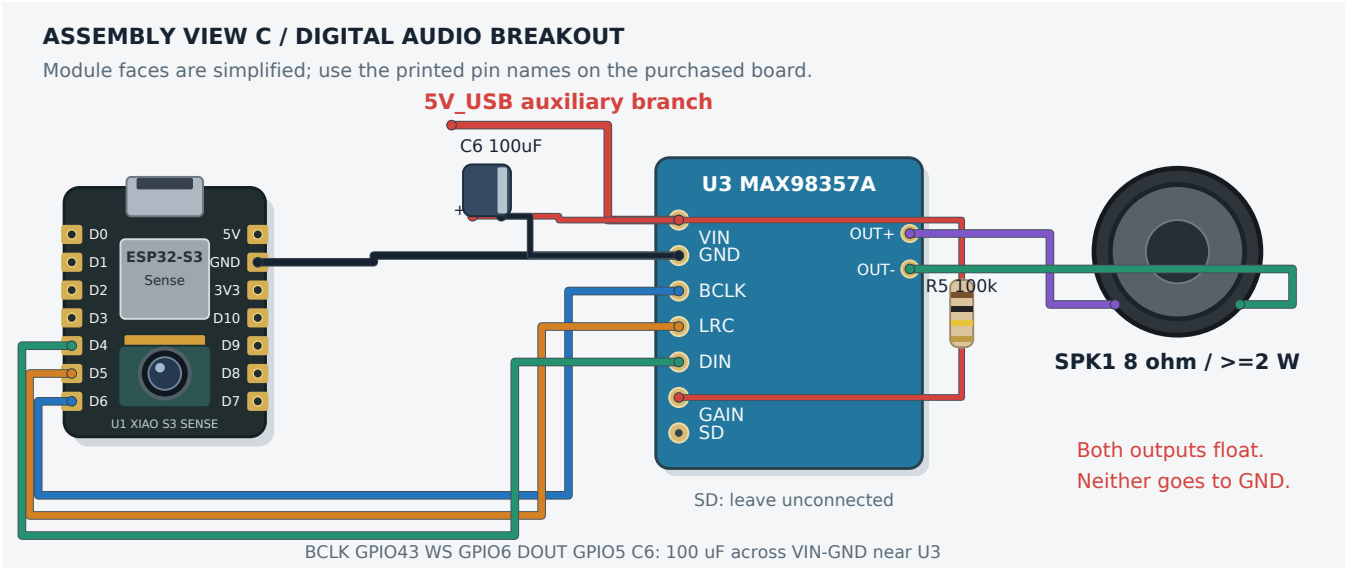


Figure 7. Illustrative module-to-module wiring. The speaker is connected only between OUT+ and OUT-. The two outputs are bridge-tied; neither is a ground connection. C6 is placed near the amplifier supply input.

Remark 3 (Clock configuration). *For 16 kHz sampling, 16-bit samples, and two slots, BCLK = 16000 x 16 x 2 = 512 kHz. Duplicate a mono sample into both slots. Use standard I2S1 TX and explicitly configure the pins; MCLK is not required [11], [19].*

The GPIO allocation avoids camera, PDM-microphone, and SD signals, but consumes D4/D5 as I2S outputs and precludes their simultaneous use as I2C. UART TX on GPIO43 must also be disabled. The present backend returns text; local playback requires an additional TTS-to-PCM transfer path.

6. Power Domains and Analytical Feasibility

Manufacturer-documented battery wiring and a transparent, assumption-based energy budget.

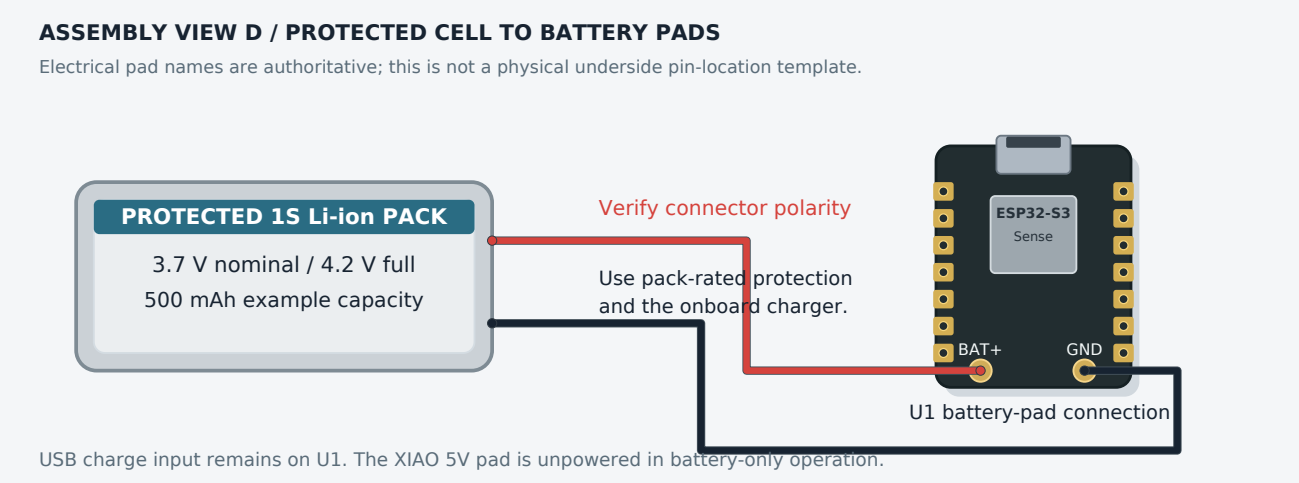


Figure 8. Protected 1S battery connected to the board battery nets. The illustration does not specify physical underside pad orientation. Match the purchased board revision and verify polarity against the manufacturer documentation [4], [6].

6.1. Power-path constraints

The referenced base-board drawing contains a 4.2 V charger programmed at approximately 110 mA and annotates a 600 mA 3.3 V regulator [6]. These values apply to that circuit revision. They do not establish maximum assembled-system demand, available peripheral headroom, or battery runtime.

In battery-only operation the XIAO 5V pad is unpowered [4]. Consequently, the USB-powered auxiliary circuits cannot be enabled by adding a battery alone. The first battery configuration should retain web-app audio; a combined battery, motor, and speaker configuration requires a separately specified supply design.

Remark 4 (Prototype power arrangement). *Use a regulated USB source for initial testing. Feed optional loads from a rated distribution branch and return their currents directly to that point. Keep the battery enclosed and off the frame during early trials; select a protected pack with suitable charge and discharge ratings.*

6.2. Illustrative runtime model

Let C be nominal capacity in ampere-hours, V the nominal cell voltage, eta an assumed usable-energy factor, and P the average core-node power. An energy estimate is:

$E_{\text{usable}} = \text{eta}\ V\ C = 0.80 \times 3.7 \times 0.5 = 1.48$ Wh. **(1)**

$t_{\text{est}} = E_{\text{usable}} / P_{\text{avg}}$. **(2)**

For a 500 mAh example pack, the resulting sensitivity to average power is:

$\mathbf{P_{avg}}$	$\mathbf{t_{est}}$
0.35 W	4.23 h
0.60 W	2.47 h
1.00 W	1.48 h

The three power values and eta are planning assumptions. They are not measurements, and phone energy is excluded. Camera revision, radio association, image size, and request frequency can all change the actual average power.

7. Reproducible Pin and Component Specification

The same references, signal names, and rail names are used in schematics and assembly views.

TABLE I. PROPOSED PIN ALLOCATION

Board label / GPIO	Assignment	Constraint
D0 / GPIO1	Capture switch	External pull-up; active-low input
D1 / GPIO2	Capture indicator	Series LED resistor required
D3 / GPIO4	Optional motor gate	100 ohm series; 100k pull-down
D4 / GPIO5	Optional I2S DOUT	Unavailable for simultaneous I2C SDA
D5 / GPIO6	Optional I2S WS	Unavailable for simultaneous I2C SCL
D6 / GPIO43	Optional I2S BCLK	Unavailable for external UART TX
GPIO41 / GPIO42	PDM microphone	Reserved DATA / CLK [9]
GPIO21 / 7 / 8 / 9	Onboard SD bus	Reserved CS / SCK / MISO / MOSI [8]

TABLE II. BILL OF MATERIALS BY CIRCUIT

Circuit	Components
A - core	U1 XIAO ESP32-S3 Sense with camera and antenna; SW1 N.O. button; LED1 green LED; R1 10k; R2 1k; C1 100nF; USB cable and regulated source.
B - haptic	U2 MCP1700-3302E/TO; Q1 AO3400A on labeled adapter; M1 Adafruit 1201 or verified equivalent; D2 1N5819; R3 100 ohm; R4 100k; C2 100nF; C3/C4 1uF; C5 47uF.
C - audio	U3 Adafruit MAX98357A mono breakout; SPK1 moving-coil 8 ohm speaker rated at least 2 W; R5 100k; C6 100uF; appropriate auxiliary 5 V distribution.
D - battery	Protected 1S pack; polarity-verified connector; insulating enclosure. The capacity and mechanical arrangement remain design choices.

The 3.3 V control rail and motor rail share ground but remain distinct regulated outputs. Auxiliary capacitors should be rated for at least 10 V, with electrolytic polarity observed. A 5 V USB source rated at 2 A is a proposed bench supply capability, not an assertion that the node continuously draws 2 A. The 250 mA motor regulator is not a substitute for the ESP32 main supply [13], [20].

8. Proposed Evaluation and Discussion

The following protocol is a plan for experiments, not a report of completed hardware measurements.

8.1. Functional and interaction protocol

Use five stationary conditions: a printed notice, a sign requiring translation, a desk scene, a simple diagram, and a deliberately unreadable image. Prepare reference answers before testing and repeat each condition under recorded lighting and distance. Log the camera identity, board revision, browser, image resolution, action, prompt, and model ID.

For each interaction, record capture time, upload completion, model response, and first spoken output. Inspect whether the answer is supported by visible evidence and whether an unanswerable image triggers useful recapture guidance. Evaluate replay and stop controls separately from answer quality.

8.2. Electrical and mechanical protocol

Measure current during acquisition, radio activity, and optional actuation. Observe supply voltage at the board, reset events, upload failures, and temperatures during a sustained bench run. Check button bounce, web-app pairing, and duplicate results. Conduct initial fit trials seated, with informed participation and supervision.

9. Discussion and Remarks

Remark 5 (Evidence boundary). *Document analysis can identify pin conflicts and unsupported power assumptions, but it cannot establish image accuracy, runtime, acoustic safety, or usability. The calculations in Section 6 are analytical estimates; the assembly illustrations are not circuit-simulator results.*

The design depends on network access and a compatible web browser. A camera may not resolve small nearby text, especially with a fixed-focus module. Current camera shipments can differ, so procurement and firmware must match [4]. Longer generated answers also increase the opportunity for unsupported detail [2].

10. Conclusion

OPTIMA will combine the glasses model with a compatible web app for deliberate capture, task selection, and spoken results. The design prioritizes a working core node, explicit power domains, and recoverable feedback. The next step is integrated bench validation of the glasses, session bridge, and web app, followed by evaluation with intended users. Haptics, local audio, and a worn battery should be added after their individual constraints have been verified.

TABLE III. MEASUREMENTS TO REPORT AFTER VALIDATION

Measure	Record / interpretation
Answer support	Correct, partially supported, unsupported, or appropriate abstention; retain the source image.
End-to-end latency	Capture to first spoken answer; report repetitions and distribution, not one best case.
Electrical reliability	Current, supply minimum, resets, upload failures, and motor-start behavior.
Interaction quality	Framing effort, speech clarity, stop/replay operation, comfort, and observed failure recovery.

Figures 3, 5, 7, and 8 are original, part-oriented vector assembly illustrations in a breadboard CAD style. They are not screenshots of a Tinkercad project and no Tinkercad simulation is claimed. Named nets and the electrical schematics define the connections.

References

Primary research and manufacturer documentation; online sources accessed 24 October 2026.

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Source: docs.espressif.com

Appendix A. Assembly Connection Schedule

Terminal references for the breadboard representation and named auxiliary carrier nodes.

In Figure 3, a-e at a given numbered position form one tie group, and f-j form a second group separated by the center trench. P/N denote the upper positive/negative rails; p/n denote the lower rails. Rails are illustrated as continuous; verify the actual breadboard with a continuity meter.

TABLE A1. CORE NODE TERMINAL SCHEDULE

Element / wire	Terminals	Electrical role
U1 3V3 / GND	P2 / N2	3.3 V control supply and ground
U1 D0 / GPIO1	a10	Capture node
R1 10k	b10 to P10	Capture pull-up
C1 100nF	c10 to c14	Capture to grounded group 14
Ground jumper	d14 to N14	Grounds group a-e / 14
Capture jumper	d10 to a18	Links capture groups 10 and 18
SW1, 2 terminals	e18 to f18	N.O. switch across the trench
Switch return	j18 to n18	Ground side of SW1
U1 D1 / GPIO2	j21	LED drive group
R2 1k	g21 to g24	Series LED current limiter
LED1 A / K	f24 / f27	Anode to resistor; cathode return
LED return / rail bridge	j27 to n27; N29 to n29	Grounds cathode and joins ground rails

A.1. Auxiliary carrier interpretation

Figure 5 uses functional carrier nodes rather than a copper artwork. U2 is shown with the TO-92 flat face forward: pin 1 GND, pin 2 VIN, pin 3 VOUT [13]. The Q1 adapter must expose labeled G, D, and S; its header arrangement is not the bare SOT-23 pad order. R3 connects GPIO4 to G, R4 connects G to S, S goes to ground, and D goes to motor negative.

The motor positive, D2 cathode, C2, and regulator output use 3V3_MOTOR. D2 anode and the other side of C2 connect to the drain. C3 decouples VIN-GND; C4/C5 decouple VOUT-GND. The control 3V3 rail remains electrically distinct from 3V3_MOTOR.

A.2. Minimum reproduction sequence

Bring up the supplied camera example over USB; verify capture and indicator control; connect the glasses to the web app through the device/session result path; then verify browser speech, replay, and stop controls. Validate that loop before adding optional circuits or moving the battery onto a wearable assembly.